REVIEW ARTICLE MANUSCRIPT DRAFT · OCTOBER 2026

Type 2 diabetes and attenuated weight loss during incretin-based therapy: mechanisms linking glucose homeostasis to energy balance

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ABSTRACT

Clinical trials of incretin-based therapies consistently report smaller reductions in body weight in people with type 2 diabetes (T2D) than in people without diabetes. We asked how robust that observation is and which mechanisms can account for it. We identified eight comparisons in which the same molecule, dose, treatment duration and estimand were tested in separate populations with and without T2D: semaglutide (two programmes), tirzepatide (two doses), orforglipron (two doses), cagrilintide–semaglutide and maridebart cafraglutide. In these comparisons the placebo-adjusted weight reduction in T2D was a median 0.62 (range 0.50–0.78) of that in populations without diabetes, an absolute difference of 2–7 percentage points. A prespecified pooled analysis of harmonised semaglutide heart-failure trials confirmed strong heterogeneity between the diabetes and non-diabetes populations ($I^2 \approx 96\%$). The deficit is graded across normoglycaemia, prediabetes and T2D, and it appears across agents with different receptor profiles. Placebo-arm weight change is not smaller in T2D, which places the deficit in the drug-attributable effect. Human studies show that incretin agonists suppress appetite in T2D, but no study has compared the magnitude of that suppression with that in people without diabetes. Correction of hyperglycaemia removes urinary glucose energy losses and lowers resting energy expenditure. Calculations based on measured inputs suggest these offsets could account for roughly 1–6 kg by 68–72 weeks, but neither has been measured in an incretin trial. Co-prescribed weight-promoting drugs, differences in age and sex, and behavioural factors probably contribute. Evidence for an intrinsic, insulin-resistance-driven resistance to incretin pharmacology is weak. The most important missing evidence is an energy-balance study, under matched conditions, that measures intake, expenditure and urinary energy losses in people with and without T2D receiving the same agent.

Keywords: incretin; GLP-1 receptor agonist; tirzepatide; semaglutide; type 2 diabetes; obesity; energy balance; glycosuria

Key points

- Across eight matched comparisons of five incretin-based molecules (same dose, duration and estimand), placebo-adjusted weight loss in type 2 diabetes was a median 0.62 times (range 0.50–0.78) that in people without diabetes: an absolute deficit of 2–7 percentage points.
- The deficit is graded across normoglycaemia, prediabetes and diabetes, appears with GLP-1, GIP/GLP-1, amylin/GLP-1 and GIP-antagonist/GLP-1 agents, and lies in the drug-attributable component, not the placebo response.
- Appetite suppression by incretin agonists is preserved in type 2 diabetes, but its magnitude has never been compared with that in people without diabetes.
- Correcting hyperglycaemia returns urinary glucose energy and lowers resting energy expenditure; assumption-driven estimates indicate these drug-arm-specific offsets could account for about 1–6 kg by 68–72 weeks, but neither has been measured in an incretin trial.
- Impaired GIP signalling or intrinsic insulin-resistance-driven pharmacological resistance are the least supported explanations; a matched energy-balance study is the key missing experiment.

1. Introduction

Medicines that act on the glucagon-like peptide-1 (GLP-1) receptor were first developed to lower glucose. They amplify glucose-dependent insulin secretion and suppress glucagon. They later proved to be the most effective drug treatments for obesity, and newer molecules add activity at the glucose-dependent insulinotropic polypeptide (GIP), glucagon or amylin receptors[1,2]. The same drugs are therefore used for two related but distinct goals: correcting dysglycaemia and reducing adiposity.

A consistent observation accompanies this dual use. When one molecule is studied at the same dose and for the same duration in separate trials of people with and without T2D, the trials in T2D report smaller weight reductions. For semaglutide 2.4 mg the placebo-adjusted reduction at 68 weeks was 12.4 percentage points without diabetes[3] and 6.2 points with T2D[4]. Similar differences appear with tirzepatide[5,6], with the oral small-molecule agonist orforglipron[7,8] and with cagrilintide–semaglutide[9,10].

The difference matters clinically, because weight reduction of this size changes glycaemic, cardiovascular and functional outcomes in T2D, and smaller responses shape both expectations and choice of agent. It is also informative mechanistically. Glucose lowering and weight reduction are separate pharmacological outputs. Correcting hyperglycaemia depends mainly on islet effects, whereas weight reduction depends on a sustained negative energy balance. The mechanisms that correct dysglycaemia need not determine weight loss, and correcting dysglycaemia can itself change energy balance.

This review addresses two questions separately. The clinical question is how confidently we can conclude that T2D is associated with less weight loss on comparable incretin-based therapy, and how large the difference is. The mechanistic question is which physiological or pharmacological processes could explain the difference, and which have actually been shown to do so. We consider four classes of explanation: (1) direct modification by T2D of the response to receptor agonism; (2) indirect metabolic effects of glycaemic correction on energy balance; (3) clinical and behavioural factors, including co-prescribed medicines; and (4) methodological artefacts arising from comparisons between trials. Throughout, we distinguish established causal links, supportive human evidence, indirect inference and hypotheses.

The evidence was assembled through structured searches of PubMed/MEDLINE and ClinicalTrials.gov, with backward citation tracking from pivotal trials. Every quantitative claim was checked against the primary abstract or, where accessible, the full text. This is a narrative review; it is not a systematic review and makes no claim of exhaustive screening (see Supplementary Methods and the quality-control report).

2. Clinical evidence: is weight loss attenuated in T2D?

2.1 What kind of comparison can answer the question?

Three kinds of evidence bear on the question, and they differ in strength. The strongest would be a randomised trial that stratifies by diabetes status and formally tests whether the treatment effect differs between strata (effect modification). No such trial has been designed for weight as its primary question. The next best are comparisons in which one protocol, or harmonised protocols, enrolled both populations, ideally analysed together from individual participant data (IPD). Weakest, but most plentiful, are comparisons of separate trials of the same molecule, dose, duration and estimand within one development programme. Comparisons of mean weight change between unrelated trials cannot distinguish an effect of diabetes from differences in population, background therapy or trial conduct. They also cannot show that individual patients respond differently, as opposed to populations differing on average.

2.2 Matched programmes

Table 1 summarises the matched comparisons, and Figure 1A shows them graphically. All use treatment-policy or treatment-regimen estimands, which count outcomes regardless of discontinuation, so the comparisons are not distorted by differences in persistence.

For semaglutide 2.4 mg weekly over 68 weeks, STEP 1 (no diabetes; baseline weight 105.3 kg, BMI 37.9 kg/m², HbA1c 5.7%) reported −14.9% versus −2.4% with placebo, an estimated treatment difference (ETD) of −12.4 percentage points[3]. STEP 2 (T2D; HbA1c 7–10%) reported −9.6% versus −3.4%, an ETD of −6.2 points[4]. For tirzepatide over 72 weeks, SURMOUNT-1 (no diabetes) reported −19.5% at 10 mg and −20.9% at 15 mg versus −3.1% with placebo[5]. SURMOUNT-2 (T2D; mean HbA1c 8.02%, diabetes duration 8.5 years) reported −12.8% and −14.7% versus −3.2%[6]. For orforglipron 36 mg over 72 weeks the results were −11.2% versus −2.1% without diabetes[7] and −9.6% versus −2.5% with T2D[8]. For cagrilintide–semaglutide over 68 weeks they were −20.4% versus −3.0% (ETD −17.3)[9] and −13.7% versus −3.4% (ETD −10.4)[10]. Liraglutide 3.0 mg showed the same direction: a placebo-adjusted −5.6 kg without diabetes[11] and −4.2 kg with T2D[12], from similar baseline weights (about 106 kg).

Several other programmes point the same way but cannot be matched as cleanly. With retatrutide, a GIP/GLP-1/glucagon receptor tri-agonist, phase 2 weight loss reached 24.2% at 48 weeks in obesity[13] and 16.9% at 36 weeks in T2D[14]. The durations differ, so the comparison is only qualitative. Oral semaglutide 50 mg produced a 15.1% reduction (ETD −12.7) in people without diabetes[15]. The corresponding T2D trial of 25 and 50 mg had no placebo arm[16]. In the SURPASS and SUSTAIN programmes in T2D, doses were chosen for glycaemic control and comparators were active. Weight reductions were 7.0–9.5 kg with tirzepatide monotherapy over 40 weeks[17], 1.9–5.5 kg greater with tirzepatide than with semaglutide 1 mg[18], and only 0.9 kg greater with semaglutide 2.0 mg than with 1.0 mg[19]. These T2D trials show that weight response rises with dose and potency in diabetes. They do not provide a matched comparator without diabetes.

Across the eight percentage-based comparisons, the placebo-adjusted reduction in T2D was a median 0.62 times that without diabetes (range 0.50–0.78). The absolute deficit was a median 5.2 percentage points (range 1.8–6.9) (Table 1; analysis/derived.py). Two features stand out. First, the ratio is broadly similar for selective GLP-1 receptor agonists, the GIP/GLP-1 co-agonist tirzepatide and the amylin–GLP-1 combination. Second, placebo-arm weight change was not smaller in T2D (−3.4% vs −2.4% for semaglutide; −3.2% vs −3.1% for tirzepatide; −2.5% vs −2.1% for orforglipron; −3.4% vs −3.0% for cagrilintide–semaglutide). The deficit therefore lies in the drug-attributable component rather than in the response to the lifestyle programme delivered to both arms.

2.3 Harmonised and single-protocol evidence

The STEP-HFpEF programme provides the closest available approach to a direct test. Two trials with harmonised eligibility criteria, endpoints and analyses enrolled people with obesity-related heart failure with preserved ejection fraction, one without diabetes and one with T2D. Both gave semaglutide 2.4 mg for 52 weeks. Weight change was −13.3% versus −2.6% without diabetes (ETD −10.7)[20] and −9.8% versus −3.4% with T2D (ETD −6.4)[21]. The prespecified IPD pooled analysis found substantial heterogeneity between the trials for weight ($p<0.0001$; I^2 95.8%), whereas improvements in heart-failure symptoms were consistent[22]. The authors note roughly 40% less weight loss with diabetes. Even here, diabetes status was not randomised, and the trials differed in recruitment and background therapy.

The phase 2 trial of maridebart cafraglutide, which combines GLP-1 receptor agonism with GIP receptor antagonism, enrolled an obesity cohort and an obesity-with-T2D cohort under one protocol[23]. At 52 weeks, weight change ranged from −12.3% to −16.2% (placebo −2.5%) in the obesity cohort and from −8.4% to −12.3% (placebo −1.7%) in the T2D cohort. The dosing arms were not identical between cohorts, and the T2D cohort was older (55.1 vs 47.9 years) and included fewer women (42% vs 63%).

Observational data point the same way. In a referral-centre cohort treated with semaglutide 1.7–2.4 mg, people with T2D lost 7.2% versus 11.8% at 6 months[24]. In a large propensity-matched electronic-record cohort, people without T2D lost more weight than people with T2D on both tirzepatide and semaglutide. The authors explicitly raise differences in motivation and concurrent weight-loss activity as possible explanations[25].

2.4 A gradient with dysglycaemia

In a post hoc analysis of STEP 1, participants with prediabetes lost 13.7% with semaglutide (ETD −11.3)[26], compared with 14.9% in the whole semaglutide arm[3]. Assuming the overall mean is the weighted average of the subgroup means, normoglycaemic participants lost about 15.9%. Normoglycaemia (~15.9%), prediabetes (13.7%) and T2D (9.6%) therefore form a graded series on the same dose and duration (Figure 1B). The gradient is unadjusted and comes from different trials, so it does not establish dose–response causation. It does argue against a threshold effect confined to overt diabetes. Its presence in prediabetes, where glycosuria is negligible, also limits how much of the gap any glycosuria-based explanation alone can carry (Section 8). In SURMOUNT-1, participants with prediabetes still lost 18.7–19.7% with tirzepatide 10–15 mg over 176 weeks[27], so large responses remain achievable in dysglycaemia.

2.5 Could the difference be an artefact?

Several features of the trials argue against a purely methodological explanation.

- **Dose, duration and estimand** were matched within each programme.
- **Discontinuation and tolerability** do not favour the trials without diabetes. Gastrointestinal adverse events were reported by 72.5% in REDEFINE 2 versus 79.6% in REDEFINE 1[9,10]. Discontinuations for adverse events were similar in ATTAIN-1 and ATTAIN-2 (5.3–10.3% vs 6.1–9.9%)[7,8].
- **Analyses that remove discontinuation** leave the difference in place. Under the efficacy estimand, which estimates the effect had treatment continued as intended, SURMOUNT-2 reported ETDs of −10.1 and −12.4 points[6]. These remain well below the treatment-regimen ETDs of −16.4 and −17.8 points in SURMOUNT-1[5], although the treatment-regimen estimand includes the effect of discontinuation and so understates the drug's full on-treatment effect.
- **Drug exposure** is unlikely to explain the difference. Within T2D programmes, population pharmacokinetic analyses identify body weight as the only clinically relevant covariate of semaglutide exposure[28], and weight and HbA1c responses track plasma exposure[29]. Baseline body weights in the matched programmes were similar. An exposure–response model of semaglutide spanning populations with and without T2D used baseline HbA1c as a covariate[30], but we could not verify from the accessible text how much it changed the predicted response.

Other differences do favour the trials without diabetes. The T2D trials enrol older participants with a higher proportion of men. Women lost more weight in the STEP-HFpEF pool[22]; STEP 1 enrolled 74.1% women and SURMOUNT-2 50.7%[3,6]. T2D participants also continued glucose-lowering drugs, including sulfonylureas in 26.7% of SURMOUNT-2 participants[6]. No analysis has pooled IPD across programmes with adjustment for age, sex, baseline BMI, HbA1c, diabetes duration and background therapy. Without one, the share of the gap attributable to these population differences cannot be quantified.

2.6 What the clinical evidence establishes

The clinical phenotype is robust. It appears in every matched programme, in harmonised IPD, in a single-protocol trial and in observational cohorts, and it is graded with glycaemic status. On average, people with T2D achieve about 50–80% (median ~60%) of the placebo-adjusted weight reduction seen without diabetes. The absolute shortfall is about 2–7 percentage points at 52–72 weeks, larger with the most potent agents. Because the deficit occurs in the drug-attributable component, and across GLP-1 receptor agonists, GIP receptor agonism and antagonism, amylin co-agonism and lifestyle intervention[31], it is more likely to reflect host physiology and context than a defect specific to one receptor. What the evidence cannot yet establish is how much of the population difference is effect modification by diabetes itself rather than confounding by co-varying traits.

Table 1. Matched comparisons of incretin-based therapies in populations with and without T2D. Values are mean % change in body weight from baseline to end of treatment (drug vs placebo). ETD = estimated treatment difference (percentage points). TP = treatment-policy estimand; TR = treatment-regimen estimand.

Agent and dose	Weeks	Population without diabetes (trial; key baseline)	Drug vs placebo; ETD	Population with T2D (trial; key baseline; background therapy)	Drug vs placebo; ETD	Ratio T2D/no-T2D (ETD)
Semaglutide 2.4 mg weekly	68	STEP 1, n=1961; 105.3 kg, BMI 37.9, HbA1c 5.7%, 74% women[3]	−14.9 vs −2.4; −12.4 (−13.4 to −11.5), TP	STEP 2, n=1210; HbA1c 7–10%; 0–3 oral agents[4]	−9.6 vs −3.4; −6.2 (−7.3 to −5.2), TP	0.50
Semaglutide 2.4 mg weekly (HFpEF)	52	STEP-HFpEF, n=529[20]	−13.3 vs −2.6; −10.7 (−11.9 to −9.4)	STEP-HFpEF DM, n=616; HbA1c ≤10%[21]	−9.8 vs −3.4; −6.4 (−7.6 to −5.2)	0.60 (IPD heterogeneity p<0.0001)[22]
Tirzepatide 10 mg weekly	72	SURMOUNT-1, n=2539; 104.8 kg, BMI 38.0[5]	−19.5 vs −3.1; −16.4*, TR	SURMOUNT-2, n=938; 100.7 kg, BMI 36.1, HbA1c 8.02%, duration 8.5 y; metformin 89%, SU 27%, SGLT2i 20%[6]	−12.8 vs −3.2; −9.6 (−11.1 to −8.1), TR	0.59
Tirzepatide 15 mg weekly	72	SURMOUNT-1[5]	−20.9 vs −3.1; −17.8*, TR	SURMOUNT-2[6]	−14.7 vs −3.2; −11.6 (−13.0 to −10.1), TR	0.65
Orforglipron 36 mg daily (oral)	72	ATTAIN-1, n=3127[7]	−11.2 vs −2.1; −9.1*, TR	ATTAIN-2, n=1613; 101.4 kg, BMI 35.6, HbA1c 8.05%[8]	−9.6 vs −2.5; −7.1 (−8.2 to −6.1), TR	0.78
Orforglipron 12 mg daily (oral)	72	ATTAIN-1[7]	−8.4 vs −2.1; −6.3*, TR	ATTAIN-2[8]	−7.0 vs −2.5; −4.5, TR	0.71
Cagrilintide 2.4 mg + semaglutide 2.4 mg weekly	68	REDEFINE 1, n=3417[9]	−20.4 vs −3.0; −17.3 (−18.1 to −16.6), TP	REDEFINE 2, n=1206; HbA1c 7–10%[10]	−13.7 vs −3.4; −10.4 (−11.2 to −9.5), TP	0.60
Maridebart cafraglutide (highest-response arm)	52	Obesity cohort, n=465; BMI 37.9, 63% women[23]	−16.2 vs −2.5; −13.7*, TP	Obesity–T2D cohort, n=127; BMI 36.5, 42% women[23]	−12.3 vs −1.7; −10.6*, TP	0.77 (dose arms differ)
Liraglutide 3.0 mg daily	56	SCALE Obesity & Prediabetes, n=3731; 106.2 kg[11]	−8.4 vs −2.8 kg; −5.6 kg	SCALE Diabetes, n=846; 0–3 oral agents[12]	−6.0% (−6.4 kg) vs −2.0% (−2.2 kg); −4.0 (−5.1 to −2.9)	0.75 (kg basis)

*ETD calculated as drug minus placebo from published means; confidence interval not reported in the abstract. SU, sulfonylurea; SGLT2i, sodium–glucose co-transporter 2 inhibitor. Ratios computed in analysis/derived.py.

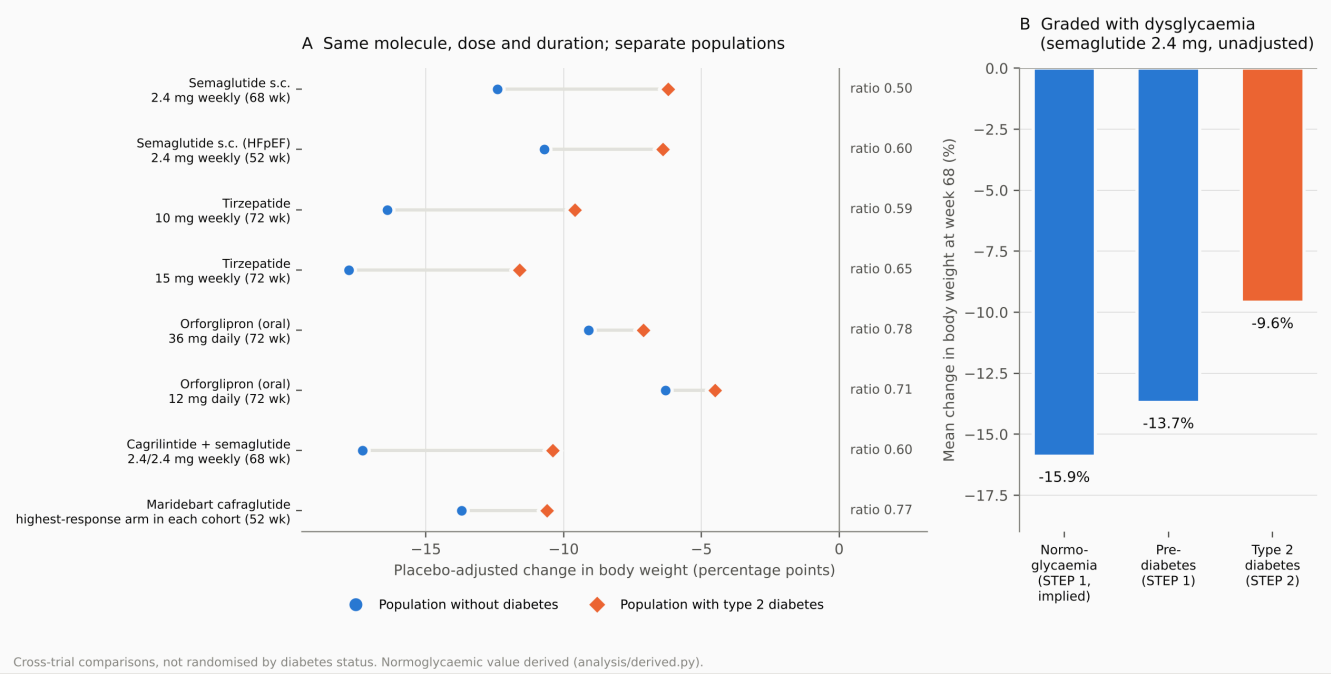


Fig. 1 | The clinical phenotype. (A) Placebo-adjusted change in body weight (estimated treatment difference, percentage points) for the same molecule, dose and duration in separate populations without diabetes (circles) and with T2D (diamonds). Values on the right are the ratio of the T2D to the non-diabetes effect. Sources are in Table 1. Comparisons are between trials and not randomised by diabetes status; the maridebart cafraglutide cohorts used different dose arms. (B) Mean weight change with semaglutide 2.4 mg at 68 weeks: normoglycaemia (implied from STEP 1 overall and prediabetes data; derived), prediabetes (STEP 1 post hoc subgroup) and T2D (STEP 2). Unadjusted cross-trial gradient. Source code: figures/make_figures.py.

3. Incretin pharmacology: separating glucose lowering from weight loss

3.1 Endogenous physiology versus pharmacology

Endogenous GLP-1 and GIP are short-lived peptides secreted after meals. Both potentiate glucose-stimulated insulin secretion. GLP-1 also suppresses glucagon, slows gastric emptying and reduces food intake[1]. In T2D the incretin effect, the extra insulin secreted after oral compared with intravenous glucose at matched glycaemia, is markedly reduced: incretin factors accounted for 36% of the insulin response in T2D versus 73% in matched controls[32]. The defect is largely a consequence of the diabetic state rather than its cause. It appears in diabetes secondary to chronic pancreatitis but not in pancreatitis with normal glucose tolerance[33]. Four weeks of near-normoglycaemia improved the late-phase β-cell response to GLP-1 and GIP three- to fourfold[34].

These findings concern the endocrine pancreas at physiological hormone concentrations. They do not imply reduced responsiveness to long-acting agonists at pharmacological exposures. Infused GLP-1 retained most of its insulinotropic activity in mild T2D, whereas the insulinotropic effect of GIP was reduced by 54%[35]. GIP also failed to amplify late-phase secretion[36]. Graded GLP-1 infusions increased β-cell glucose responsiveness in T2D, but at each dose the response was 3–5 times smaller than in controls. Even so, a low dose restored responsiveness to the level of untreated controls[37]. A reduced physiological incretin effect is therefore a statement about the islet. It says nothing direct about the brain and gut circuits through which agonists reduce energy intake.

3.2 Two outputs, partly separate pathways

The glucose-lowering effects of incretin drugs arise mainly from islet mechanisms (insulin secretion and glucagon suppression), from slower gastric emptying of meal glucose, and secondarily from weight reduction and improved insulin sensitivity. In a mechanistic study in T2D, tirzepatide improved the clamp disposition index, total insulin secretion, insulin sensitivity (M value) and glucagon suppression more than semaglutide 1 mg[38]. Weight reduction, by contrast, depends on a sustained negative energy balance. That balance is driven mainly by reduced energy intake, with little or no increase in energy expenditure in humans (Sections 4 and 9). Both outputs share the receptor but not the effector organs. This is why a drug can improve glycaemia without changing weight, as low-dose liraglutide (0.6 mg) did in an 8-week respiratory-chamber study in T2D[39], and why glycaemic correction can feed back on energy balance (Section 8).

3.3 Exposure, receptor engagement and molecular class

Long-acting peptide agonists achieve sustained plasma concentrations far above endogenous GLP-1. Within T2D, semaglutide exposure varies mainly with body weight[28]. Oral and subcutaneous formulations show nearly identical exposure–response relationships for HbA1c and body weight, which suggests that variation between individuals in GLP-1 receptor responsiveness is not a major determinant of response[29]. Agents differ in other ways. Tirzepatide engages the GIP receptor more than the GLP-1 receptor at clinical doses, and at the GLP-1 receptor it favours cAMP signalling over β-arrestin recruitment and causes less receptor internalisation[40]. Orforglipron is a non-peptide oral agonist[7]. Cagrilintide adds amylin receptor agonism[9]. Maridebart cafraglutide pairs GLP-1 agonism with GIP receptor antagonism[23]. Despite these differences, the attenuation ratio in T2D is similar across agents (Table 1). That similarity is the first indication that the explanation lies mainly in the host rather than in a receptor-specific pharmacodynamic defect.

4. Central regulation of appetite and energy intake

4.1 How incretin agonists reduce intake

Reduced energy intake is the dominant driver of incretin-induced weight loss. In adults with obesity, semaglutide 1.0 mg reduced ad libitum energy intake across a test day by 24%, with less hunger and food craving, better control of eating and a lower preference for high-fat foods. Resting metabolic rate adjusted for lean mass did not change[41]. Semaglutide 2.4 mg reduced ad libitum lunch intake by 35% after 20 weeks[42]. Tirzepatide reduced intake at an ad libitum lunch by about 525 kcal at week 3 in adults without diabetes. It also reduced food cravings, the tendency to overeat and responsiveness to food cues, and reduced activation to high-fat, high-sugar food images in orbitofrontal, cingulate and hippocampal regions[43]. In humans, GLP-1 receptors have been identified on neurons of the hypothalamus, medulla and parietal cortex[44].

Animal studies define the circuitry. Peripherally administered semaglutide does not cross the blood–brain barrier. It reaches circumventricular organs and periventricular sites, activates hindbrain and hypothalamic neurons, and engages secondary nodes such as the lateral parabrachial nucleus. In rodents it reduces intake without lowering energy expenditure[45]. Liraglutide directly activates arcuate POMC/CART neurons and indirectly inhibits NPY/AgRP neurons through GABAergic transmission. Its weight-lowering effect persisted after loss of GLP-1 receptors in vagal afferents, area postrema or paraventricular nucleus[46]. Other work shows that vagal afferent and central receptor populations both contribute, because vagal deafferentation shifts the dose–response curve[47]. The pharmacological effect is also separable from the endogenous brainstem preproglucagon system; those neurons are not required for the anorexia induced by GLP-1 receptor agonists[48]. These rodent data describe pathways. They do not show that any pathway behaves differently in human T2D.

4.2 Appetite suppression in T2D: present, magnitude untested

Three types of evidence should be kept apart.

Incretin agonism suppresses appetite in T2D. This is well supported. Intravenous GLP-1 reduced energy intake by 27% in T2D[49]. Liraglutide 1.8 mg reduced ad libitum intake by 18% after 3 weeks[50]. Tirzepatide and semaglutide both reduced appetite in T2D over 28 weeks[51]. In a crossover fMRI study, exenatide reduced food intake and food-cue responses in insula, amygdala, putamen and orbitofrontal cortex in obese people with T2D as well as in normoglycaemic obese people, and receptor blockade with exendin 9-39 largely prevented these effects[52]. In T2D, liraglutide reduced insula and putamen responses to food pictures after 10 days but not after 12 weeks, relative to insulin glargine[53]. It also reduced parietal activation to highly desirable foods[44].

Whether appetite suppression is smaller in T2D is unknown. No study has compared ad libitum intake, appetite ratings or food-cue responses between people with and without diabetes receiving the same agonist under identical conditions. Apparent differences between studies, such as 27% with GLP-1 infusion in T2D versus 12% in healthy men[54], reflect different infusion rates, meals and populations and cannot be interpreted as effect modification.

Whether a difference in appetite suppression explains the weight-loss gap has therefore never been tested. One observation is suggestive. In T2D, tirzepatide produced more weight loss than semaglutide 1 mg, but intake at an ad libitum lunch did not differ between the drugs. The authors concluded that measured intake did not account for the difference in weight outcome[51]. A single test meal is an insensitive measure of 24-hour intake, so this finding should not be overinterpreted.

4.3 Could the diabetic state blunt central responses?

Several plausible mechanisms exist, but none has been linked to incretin response. In adults with obesity, MRI markers of mediobasal hypothalamic gliosis were greater in impaired glucose tolerance and T2D than in normal glucose tolerance, independent of adiposity. They correlated with glycaemia and insulin resistance[55]. Brain insulin resistance in obesity and T2D alters eating behaviour and reward processing[56]. In obese T2D and normoglycaemic obese participants, food-cue responses in appetite and reward regions were greater than in lean participants[52]. A larger baseline drive to eat would oppose a given pharmacological suppression. Whether gliosis, central insulin resistance or heightened food-cue reactivity reduces the *magnitude* of GLP-1-induced anorexia in T2D has not been studied. These remain hypotheses.

4.4 Compensatory regulation

Weight loss provokes compensatory increases in appetite. In patients treated with canagliflozin, whose energy deficit comes from glycosuria without appetite suppression, calculated energy intake rose by about 100 kcal/day for each kilogram lost. That is more than three times the accompanying adaptation in energy expenditure[57]. A common interpretation, not directly tested in humans, is that incretin agonists shift the level at which this feedback operates. Under that model, any additional, drug-independent energy inflow in T2D, such as glucose energy previously lost in urine (Section 8), would act on this feedback system and lead to a higher weight plateau.

5. Gastrointestinal physiology and gastric emptying

GLP-1 slows gastric emptying acutely. With sustained receptor exposure this effect diminishes. Tachyphylaxis of the delay was evident by a second meal during an 8.5-hour GLP-1 infusion, together with loss of the associated vagal marker pancreatic polypeptide[58]. A continuous 24-hour infusion attenuated the delay, whereas intermittent exposure preserved it[59]. With long-acting agents the picture is mixed. After 20 weeks of semaglutide 2.4 mg, paracetamol absorption showed no delay[42]. Semaglutide 1.0 mg delayed only first-hour emptying[60]. Exenatide once weekly still slowed scintigraphically measured emptying of solids and liquids at 8 weeks[61]. Liraglutide 3.0 mg delayed solid emptying at 5 and 16 weeks in obesity, and the delay at 5 weeks correlated with weight loss at 16 weeks[62]. Tirzepatide's delay of gastric emptying likewise waned with repeated dosing in healthy participants. In T2D, however, a residual delay persisted during dose escalation, and a long-acting GIP analogue alone did not affect emptying[63].

Diabetes affects this axis in ways that could push in either direction. Acute hyperglycaemia slows gastric emptying, as shown in type 1 diabetes under glucose clamp[64]. Diabetic autonomic neuropathy can cause gastroparesis[65]. The glucose-lowering effect of GLP-1 receptor agonism through slower gastric emptying depends on the baseline

emptying rate[66,67]. If emptying is already slow in some people with T2D, the incremental slowing a drug can produce, and any satiation it adds, may be smaller. Conversely, correcting hyperglycaemia could speed emptying, partly offsetting the drug effect. Neither possibility has been tested against weight outcomes.

The weight of evidence suggests that gastric emptying contributes to early satiation and perhaps to variation between individuals in response[62]. It is not the principal mechanism of long-term weight loss, because weight loss continues after the delay has largely waned, and central receptor populations are sufficient for anorexia in animal models[45,46]. Gastrointestinal changes in T2D are therefore best regarded as a variable and probably minor modifier. They would need to be measured, together with weight, in both populations before they could be assigned any share of the gap. Tolerability, the clinically visible face of gut effects, does not appear worse in T2D trials (Section 2.5).

6. GIP biology and dual receptor agonism

6.1 Established human pharmacology

The insulinotropic action of GIP is reduced in T2D, both in peak effect[35] and in amplifying late-phase secretion[36]. Near-normalisation of glycaemia partly restores it[34]. In rat islets, chronic hyperglycaemia reduces expression of both GLP-1 and GIP receptors, and normalising glucose reverses the change[68]. This finding suggests a mechanism, but it is animal data. GIP has adipose actions in humans. Together with hyperinsulinaemia it increased subcutaneous adipose blood flow, glucose uptake, triglyceride hydrolysis and fatty-acid re-esterification[69]. Infused GIP did not reduce energy intake in overweight men, and adding GIP to GLP-1 did not enhance the GLP-1-induced reduction in intake[70].

6.2 Mechanisms proposed mainly from animal and cellular studies

In mice, hypothalamic cells expressing the GIP receptor suppress food intake when activated[71]. Central GIP receptors are required for the anorectic effect of acyl-GIP and for the greater efficacy of GLP-1/GIP co-agonism over GLP-1 alone[72]. In mice, rats and shrews, GIP receptor agonism attenuates GLP-1-induced emesis while preserving reduced food intake[73]. Paradoxically, GIP receptor antagonist antibodies also reduce weight and enhance GLP-1 agonist effects in rodents and non-human primates[74]. One explanation is that sustained agonism produces functional antagonism through receptor desensitisation, but this remains unresolved. At the receptor level, tirzepatide is biased in its signalling and engages the GIP receptor more than the GLP-1 receptor[40].

6.3 Does altered GIP physiology in T2D predict a different response to dual agonism?

If reduced GIP receptor responsiveness in T2D blunted the extra weight loss attributable to GIP agonism, the ratio of T2D to non-T2D weight loss should be lower for tirzepatide than for selective GLP-1 receptor agonists. It is not: the ratio was 0.59–0.65 for tirzepatide and 0.50–0.60 for semaglutide (Table 1). The attenuation is also present with maridebart cafraglutide, which *antagonises* the GIP receptor (ratio about 0.77)[23], and with orforglipron, which has no GIP activity[8]. The weight deficit in T2D therefore does not depend on the direction of GIP receptor pharmacology. This makes impaired GIP signalling an unlikely primary explanation.

The mechanistic data in T2D point instead to effects of tirzepatide beyond intake. In T2D, tirzepatide produced more weight and fat-mass loss than semaglutide 1 mg without a measurable difference in ad libitum intake[51]. It also produced larger improvements in insulin sensitivity[38]. In people with obesity but without diabetes, tirzepatide increased fat oxidation without changing metabolic adaptation[75]. Larger weight loss with dual agonism does not prove a specific GIP-mediated anorectic mechanism in humans. The relative contributions of greater GLP-1 receptor exposure, GIP-mediated tolerability, adipose effects and substrate effects are not established.

7. Insulin resistance, hyperinsulinaemia and substrate partitioning

7.1 The pathophysiological premise

Skeletal muscle is the main site of insulin-stimulated glucose disposal[76]. In T2D, insulin resistance affects muscle, liver and adipose tissue. Hepatic insulin resistance sustains glucose production, adipose insulin resistance weakens insulin's suppression of lipolysis, and ectopic lipid accumulation links these defects[77]. Insulin-resistant muscle shows "metabolic inflexibility": it oxidises more glucose in the fasting state but fails to switch towards glucose oxidation when stimulated by insulin[78]. Incretin therapy improves several of these defects. Tirzepatide increased the clamp M value in T2D[38]. In patients with non-alcoholic steatohepatitis, liraglutide improved hepatic and adipose insulin sensitivity and reduced de novo lipogenesis[79]. Semaglutide reduced HOMA-IR in people with prediabetes[26].

7.2 Does insulin resistance plausibly limit weight loss?

A recurrent proposal is that insulin resistance itself, or the hyperinsulinaemia it causes, traps energy in adipose tissue or limits fat mobilisation and so slows weight loss. Three considerations argue against assigning it a primary role.

1. **Substrate handling does not set the energy balance.** Muscle glucose uptake, glucose oxidation and glycogen storage describe which fuel is used and where it is stored. Total energy expenditure is set by ATP demand. A shift from glucose to fat oxidation, or the reverse, does not by itself change the energy deficit. In a controlled metabolic-ward study, isocaloric carbohydrate restriction lowered insulin and raised fat oxidation, yet produced less body-fat loss than equal-calorie fat restriction[80]. Insulin-driven partitioning therefore did not override energy balance.
2. **The mechanistic link is unmeasured.** We found no human study in T2D that measured muscle glucose uptake, glucose oxidation, glycogen storage and total energy expenditure together during incretin treatment. The proposition that impaired insulin-mediated glucose disposal reduces incretin-associated weight loss is therefore untested.
3. **Treatment improves insulin sensitivity.** Insulin sensitivity improves during treatment in both populations, which would narrow any partitioning difference over time.

One partitioning effect deserves note because it changes weight without changing fat: glycogen. Glycogen is stored with water, so a larger glycogen pool can add weight. Improved insulin action could plausibly restore muscle and liver glycogen that are depleted in poorly controlled diabetes and add weight unrelated to adiposity. This is an inference from general physiology and has not been measured in incretin trials.

7.3 Endogenous hyperinsulinaemia versus insulin and sulfonylurea therapy

Endogenous hyperinsulinaemia must be distinguished from weight-promoting medicines. In UKPDS 33, intensive treatment with insulin increased weight by 4.0 kg, and sulfonylureas by 1.7–2.6 kg, relative to conventional therapy[81]. In SURMOUNT-2, 26.7% of participants took sulfonylureas, and level-2 hypoglycaemia clustered in that group despite protocol-guided dose reduction[6]. Insulin-associated weight gain arises partly through the energetics of improved glycaemia (Section 8). Hypoglycaemia-avoidant eating is often invoked but is a hypothesis. Notably, liraglutide 3.0 mg plus intensive behavioural therapy produced a placebo-adjusted reduction of 4.3 percentage points in insulin-treated T2D[82], similar to the 4.0 points in T2D without insulin[12]. Insulin co-therapy with down-titration therefore did not abolish the response. Background therapy is a credible contributor to the gap at population level, but no trial has stratified weight response by background agent with adequate power.

8. Glycosuria, glycaemic correction and net energy balance

8.1 The premise

When plasma glucose exceeds the renal reabsorptive capacity, glucose is lost in urine and its energy leaves the body. In people with T2D under stepped hyperglycaemic clamps, urinary glucose excretion rose from 0.06 mmol/min at 7.8 mmol/L to 0.77 mmol/min at 13.3 mmol/L. Glycosuria persisted down to 7.8 mmol/L, and the threshold varied between people from 7.7 to 12.2 mmol/L[83]. Restoring glycaemia closes this energy leak. The energy is then retained, acting like an increase in intake, but only in people who were glycosuric, and only in the arm whose glycaemia improves.

8.2 Measured magnitudes

Three human studies measured the energy involved directly.

- In obese Pima Indians with T2D, 6 weeks of sulfonylurea lowered fasting glucose from 253 to 144 mg/dL and urinary glucose from 128 to 11 g/day. Resting metabolic rate per kilogram of fat-free mass also fell. To keep weight constant, daily intake had to be cut from 3070 to 2784 kcal[84].
- In type 1 diabetes, intensive insulin therapy under constant intake reduced glycosuria from 428 to 39 mmol/day. Whole-room calorimetry showed a 5% (118 kcal/day) fall in 24-hour energy expenditure, and weight increased by 2.6 kg, almost all of it fat[85].
- In insulin-treated T2D, improved glycaemia was an independent determinant of lower basal metabolic rate. The authors concluded that better glycaemic control promotes weight gain by reducing both metabolic rate and glucosuria[86].

The reverse experiment confirms that urinary glucose energy matters to weight. SGLT2 inhibition with empagliflozin caused about 206 kcal/day of glycosuric loss, which predicted an 11.3 kg weight loss. Observed loss was 3.2 kg because calculated intake rose by about 13%[87]. With canagliflozin, intake rose by about 100 kcal/day for each kilogram lost[57].

8.3 Could this offset be large enough?

None of the incretin trials measured urinary glucose or energy expenditure. We therefore built transparent scenario estimates (Table 2; analysis/derived.py). Each assumes a *differential* reduction in glycosuria between drug and placebo arms, plus a differential fall in resting energy expenditure, persisting over the trial. We used 3.75 kcal per gram of glucose and the energy–weight relationship of Hall and colleagues: a permanent 100 kJ/day change yields about 1 kg at steady state, with half the change reached in about 1 year[88]. For trial-like T2D populations with baseline HbA1c of about 8%, in which the drug lowers HbA1c by about 1.5 percentage points more than placebo[6], the retained energy could account for roughly 1 kg (low scenario) to 6–7 kg (high scenario) by 68–72 weeks. The observed placebo-adjusted gap between STEP 1 and STEP 2 is about 6 percentage points, or roughly 6 kg.

These calculations show that the energetics of glycaemic correction are *quantitatively capable* of explaining a meaningful share of the gap. They do not show that they do. Four caveats apply. Baseline glycosuria in trial populations was not measured and varies widely between individuals[83]. Behavioural compensation could be blunted by the drug's own anorectic effect, or it could add to the offset. The anchoring studies of metabolic rate involved poorer control than in modern trials[84]. And the gradient seen in prediabetes, where glycosuria is negligible (Section 2.4), shows that glycosuria cannot be the whole explanation.

The mechanism does fit two features of the clinical phenotype. It operates only in the arm whose glycaemia improves, so it predicts a deficit in drug-attributable rather than placebo-arm weight change, as observed. And because people without diabetes have no glycosuria to lose, it predicts no offset in non-diabetes trials.

Table 2. Energy offsets from glycaemic correction: measured anchors and assumption-driven scenarios.

Source / scenario	Input	Energy (kcal/day)	Weight equivalent
Bogardus 1986, T2D, sulfonylurea[84]	Urinary glucose 128 → 11 g/day (measured)	~439	—
Carlson & Campbell 1993, type 1 diabetes, intensive insulin[85]	Glycosuria 428 → 39 mmol/day; 24-h EE −118 kcal/day (measured)	~263 (glycosuria) + 118 (EE)	+2.6 kg in 2 months at constant intake
Ferrannini 2015, empagliflozin[87]	Glycosuric loss (estimated from measurements)	~206	Predicted −11.3 kg; observed −3.2 kg (compensation)
Rave 2006 clamp[83]	UGE 0.06 → 0.77 mmol/min at 7.8 → 13.3 mmol/L	58–749 if sustained for 24 h	—
Scenario: low	10 g/day differential glycosuria; no REE change	38	~1.0 kg by 68–72 weeks
Scenario: central	25 g/day; REE −40 kcal/day	134	~3.3 kg by 68–72 weeks
Scenario: high	45 g/day; REE −90 kcal/day	259	~6.4–6.7 kg by 68–72 weeks

Scenarios are illustrative assumptions, not measurements. Conversion: 23.9 kcal/day per kg at steady state; exponential approach with a 1-year half-time[88].

9. Adipose tissue, body composition and metabolic adaptation

Composition of weight lost. In a DXA substudy of SURMOUNT-1 (no diabetes), tirzepatide reduced weight by 21.3%, fat mass by 33.9% and lean mass by 10.9%. About three-quarters of the weight lost was fat in both drug and placebo groups[89]. In T2D, semaglutide 1.0 mg reduced fat mass by 3.4 kg and lean mass by 2.3 kg over 52 weeks in a SUSTAIN 8 substudy, but there was no placebo arm and the dose was lower[90]. Tirzepatide reduced liver fat and visceral and subcutaneous adipose volumes compared with insulin degludec[91]. Semaglutide 2.4 mg reduced visceral fat area by 40% in an East Asian population with and without T2D[92]. No study has compared body-composition changes between populations with and without T2D at matched dose and duration.

Body composition cannot easily explain smaller weight loss. Losing a kilogram of fat removes about 39.5 MJ of stored energy, compared with about 7.6 MJ for a kilogram of lean tissue[88]. If people with T2D lost a larger share of weight as lean tissue, a given energy deficit would produce *more* weight loss, not less. A composition difference could reduce weight loss only through fluid or glycogen shifts (Section 7.2). Those are short-term effects and have not been measured in this context.

Energy expenditure and adaptation. In humans, incretin agonists do not increase energy expenditure. Semaglutide did not change resting metabolic rate adjusted for lean mass[41]. Liraglutide did not change 24-hour expenditure in T2D[39] and transiently *reduced* resting expenditure in a 26-week T2D trial[93]. Tirzepatide did not change metabolic adaptation in obesity but increased fat oxidation[75]. The GLP-1/glucagon co-agonist cotadutide produced weight loss in T2D driven mainly by reduced intake, with little change in expenditure[94]. One adaptation is specific to diabetes. Resting metabolic rate is about 5% higher in T2D than in equally obese people with normal glucose tolerance, and it falls as glycaemia is corrected[84,86]. Normalising glycaemia thus removes an energy cost unique to the diabetic state, adding to the glycosuric offset in Section 8.

Physical activity has not been measured objectively and compared between populations in incretin trials. Whether lower activity in older, more comorbid T2D populations contributes is unknown.

10. Integrative model: from diabetes pathophysiology to weight-loss response

Figure 2 organises the evidence into four interacting domains. Each causal link is labelled by the strength of its support. Table 3 summarises the evidence for each candidate mechanism.

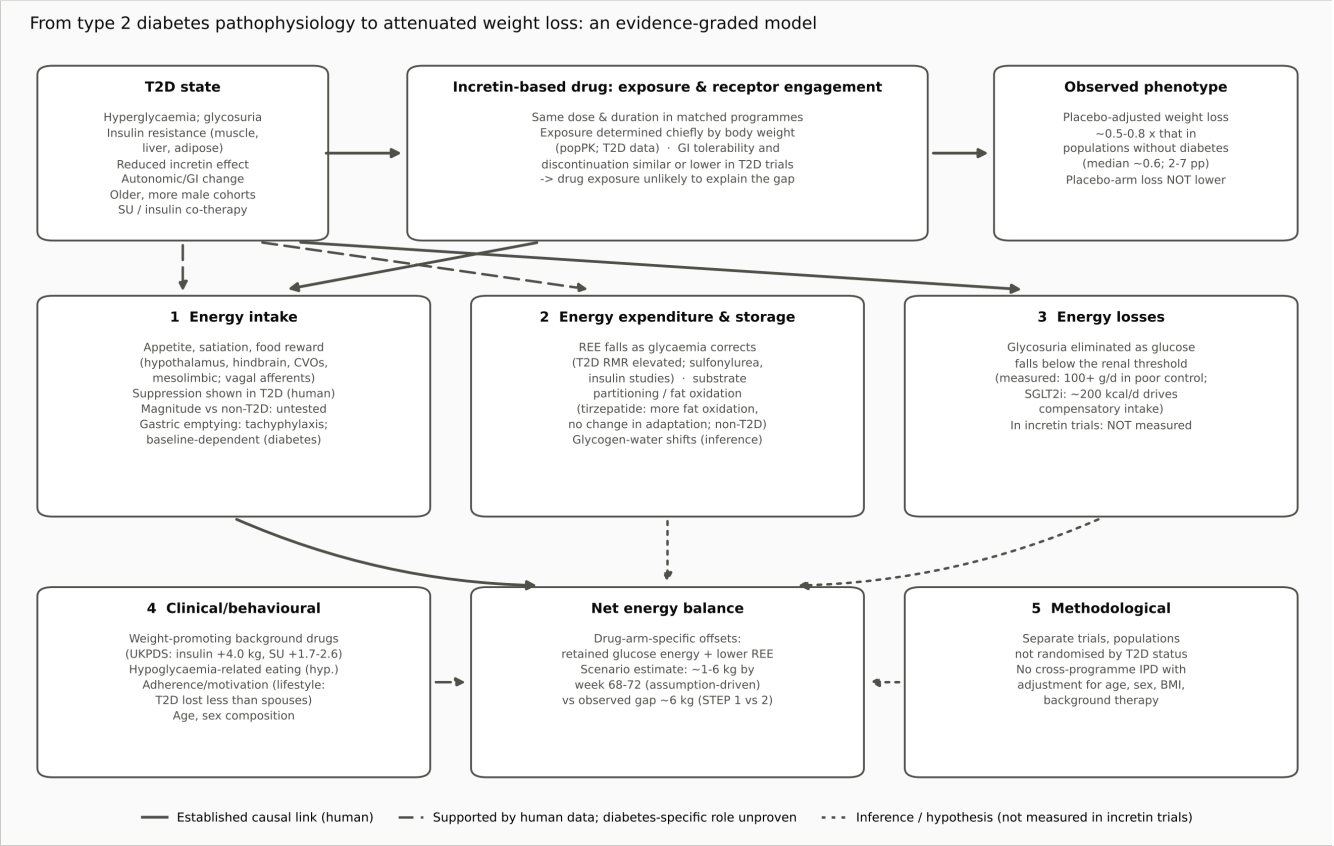


Fig. 2 | Evidence-graded integrative model. Features of T2D act on three components of energy balance (intake, expenditure and storage, and losses), alongside clinical and behavioural factors and methodological sources of between-trial difference. Solid lines denote causal links established in humans. Dashed lines denote links supported by human data whose diabetes-specific role is unproven. Dotted lines denote inferences or hypotheses not measured in incretin trials. Scenario values are assumption-driven (Table 2). The figure does not apportion the gap among mechanisms.

Drug exposure and pharmacodynamics. Dose, duration and estimand were matched in the comparisons. Exposure depends mainly on body weight, and tolerability and discontinuation do not favour the trials without diabetes. Receptor pharmacology varies widely across agents that all show the deficit, including opposite directions of GIP receptor action. Reduced drug exposure or a receptor-specific defect is therefore unlikely to be a primary determinant.

Energy intake. Intake suppression is the main mechanism of incretin-induced weight loss in both populations and is preserved, at least qualitatively, in T2D. Its magnitude has never been compared between populations under matched conditions. A smaller central or gastrointestinal effect in T2D, through hypothalamic gliosis, central insulin resistance, greater food-cue reactivity or slower baseline gastric emptying, is biologically plausible but unmeasured. This is the largest unknown in the model.

Energy expenditure and substrate metabolism. Agonists do not raise expenditure, and adaptive changes appear similar in kind across populations. Correcting hyperglycaemia removes the excess resting metabolic rate seen in T2D. Changes in substrate partitioning and insulin sensitivity alter fuel selection, but there is no evidence that they set the size of the energy deficit.

Energy losses and glycaemic correction. Elimination of glycosuria returns previously lost glucose energy to the body. The size of this effect is established in diabetes generally, and scenario estimates show it could account for a meaningful share of the gap in the drug arm. It has never been measured in an incretin trial.

How several modest effects combine. Together, the energy offsets from glycaemic correction (glycosuria and metabolic rate) and weight-promoting co-therapies could narrow the net energy deficit in the drug arm. Because weight approaches a new steady state over a year or more, small persistent offsets matter: each 24 kcal/day of persistent offset corresponds to about 1 kg at steady state[88]. Population differences in sex and age, and possibly behaviour[25,31], add further differences between trials. We deliberately do not assign percentage contributions. None of these components has been measured in both populations under the same treatment, so any apportionment would be speculative.

Which explanations are best supported? Ranked by evidence:

- **Best supported.** The phenotype exists, is graded and is confined to the drug-attributable component. Drug exposure and tolerability do not explain it.
- **Plausible and quantitatively capable, but unmeasured.** Energetic offsets from glycaemic correction (glycosuria, lower REE). Weight-promoting background therapy. Population composition by sex and age.
- **Plausible, magnitude unknown.** Attenuated appetite suppression in T2D. Gastrointestinal modifiers.
- **Weakest.** An intrinsic, insulin-resistance-driven pharmacological resistance. Impaired GIP signalling as the primary cause.

Table 3 | Mechanistic evidence map.

Mechanism (category)	Biological hypothesis	Evidence in T2D	Evidence without diabetes	Direct comparison by diabetes status	Human vs animal	Causal evidence	Quantitative contribution	Confidence
Reduced drug exposure / tolerability (clinical)	Lower exposure or more discontinuation in T2D	Exposure set by body weight[28,29]; GI events and discontinuation not higher[8,10]	Same[7,9]	Indirect (matched programmes)	Human	Argues against	Not a major contributor	Moderate (against)
Attenuated appetite suppression (direct)	Diabetic state blunts central/vagal anorectic response	Suppression present[49–52]	Suppression present[41–43]	None	Human (effect); animal (circuits)[45,46]	Not tested	Unknown	Low
Central gliosis / brain insulin resistance (direct)	Hypothalamic injury or insulin resistance lowers drug efficacy	Gliosis greater in IGT/T2D[55]; brain IR[56]	—	None	Human association	None	Unknown	Very low
Gastric emptying (direct/indirect)	Slower baseline GE or neuropathy limits incremental slowing	GE effect baseline-dependent[66,67]; residual delay with tirzepatide in T2D[63]; hyperglycaemia slows GE[64]	Delay wanes; correlates with weight loss[42,62]	None	Human	Not tested	Probably minor	Low
Impaired GIP signalling (direct)	Reduced GIPR response blunts tirzepatide-specific weight loss	GIP insulinotropic action reduced[35,36]; partly restored[34]	GIP does not reduce intake in humans[70]	Ratios similar across GIPR agonist, antagonist and GLP-1-only agents (Table 1)	Human + animal[72,74]	Argues against primary role	Small at most	Moderate (against)
Glycosuria elimination (indirect)	Retained glucose energy offsets the deficit in the drug arm	Large measured glycosuria in poor control[83,84]; SGLT2i energetics[87]	Absent	Not measured in incretin trials	Human	Established energetics; mediation untested	Scenario ~1–6 kg	Moderate (plausibility); low (mediation)
REE fall with glycaemic correction (indirect)	Excess REE of T2D removed	RMR +5% in T2D, falls with treatment[84,86]; EE −5% with intensive insulin (T1D)[85]	Not applicable	Not measured in incretin trials	Human	Established in other settings	~40–100 kcal/day (assumption)	Moderate (plausibility)
Insulin resistance / substrate partitioning (direct)	IR or hyperinsulinaemia limits fat mobilisation	IR improves on treatment[38,79]	Isocaloric partitioning does not override energy balance[80]	None	Human	None for weight	Unknown; probably small	Low
Weight-promoting co-therapy (clinical)	SU/insulin add energy retention or defensive eating	Weight gain with SU/insulin[81]; SU in 27%[6]; insulin-treated response preserved[82]	Absent	None stratified	Human	Plausible	Unknown	Low–moderate
Population composition (methodological)	Age, sex and comorbidity differ between programmes	Older, more men[6,23]	More women[3]	Sex interaction for weight[22]	Human	Confounding plausible	Unknown	Moderate (plausibility)
Behaviour / adherence (clinical)	Lower dietary adherence or motivation	T2D lost less with behavioural programme; smaller intake reduction[31]; motivation proposed[25]	—	Small direct study (n=12 pairs)	Human	Associational	Unknown	Low

11. Unresolved questions and future directions

The central gap is that no study has measured the components of energy balance in people with and without T2D receiving the same incretin agent. Table 4 sets out studies that would resolve the main uncertainties, in order of priority.

Table 4 | Proposed studies.

#	Hypothesis	Population	Key measurements	Primary endpoint	Supports the hypothesis if…	Refutes it if…
1	The gap persists after adjustment for population composition	IPD from STEP 1/2, SURMOUNT-1/2, ATTAIN-1/2, REDEFINE 1/2	Weight; age, sex, BMI, HbA1c, diabetes duration, background therapy, adherence	Adjusted diabetes × treatment interaction for % weight change	An adjusted interaction remains, graded by HbA1c	The interaction disappears after adjustment
2	Glycaemic correction offsets the energy deficit	Matched adults with obesity ± T2D starting the same agent; 26–52 weeks	24-h urinary glucose; doubly labelled water; whole-room calorimetry; DXA	Change in urinary energy loss plus REE change versus weight-loss difference	Retained energy accounts for a substantial share of the weight-loss difference	Urinary and REE changes are small (<30 kcal/day)
3	Appetite suppression is smaller in T2D	Same cohort as #2, with an early crossover arm	Ad libitum intake (24 h), appetite ratings, food-cue fMRI at weeks 3 and 20	Placebo-adjusted reduction in intake by diabetes status	Smaller reduction in T2D, predicting weight response	Equal reduction in intake
4	Insulin resistance modifies weight response	Same cohort, stratified by clamp M value	Hyperinsulinaemic–euglycaemic clamp; stable-isotope tracers of lipolysis and glucose disposal; muscle glycogen by ^{13}C-MRS	Association of baseline M value with weight response, adjusted for intake	M value predicts response independently of intake and glycosuria	No independent association
5	Background therapy contributes	T2D trial participants randomised to protocolised sulfonylurea/insulin withdrawal versus continuation	Weight; hypoglycaemia; intake	Difference in weight response	Withdrawal increases weight loss	No difference

#	Hypothesis	Population	Key measurements	Primary endpoint	Supports the hypothesis if…	Refutes it if…
6	The glycaemic gradient is causal	T2D participants on the same agent randomised to SGLT2i co-therapy (maintaining glycosuria) versus none	Urinary glucose; weight; intake	Weight difference attributable to sustained glycosuria	Weight loss in T2D approaches non-diabetes levels	No additional weight difference beyond the SGLT2i effect

Several design principles apply across these studies:

- **Mediation analyses.** Studies should prespecify mediation analyses that partition the diabetes difference into intake, expenditure and urinary-loss components, rather than reporting each in isolation.
- **Controlled feeding.** Short controlled-feeding studies under matched agonist exposure could separate drug effects on intake from effects on energy disposal.
- **Measurement in T2D trials.** Phase 3 programmes in T2D could add urinary glucose collection and expenditure measurement at low cost, which would make these data routine.

12. Conclusions

Clinical question. The evidence firmly supports an association between T2D and less weight loss on comparable incretin-based therapy. Across matched programmes of five molecules, the placebo-adjusted reduction in T2D was about 50–80% (median ~60%) of that without diabetes, an absolute shortfall of about 2–7 percentage points at 52–72 weeks. The finding is consistent across programmes, in harmonised IPD and in observational cohorts, and it is graded with glycaemic status. Certainty that the *association* exists is high. Certainty that it reflects *effect modification by diabetes itself*, rather than co-varying population traits, is moderate at best, because no adjusted cross-programme IPD analysis exists.

Mechanistic question. No mechanism has been shown to explain the gap. The best-supported explanations involve the energy balance of the drug-treated person with diabetes rather than resistance to the drug. Correcting hyperglycaemia returns glycosuric energy and removes the excess resting metabolic rate of T2D. These effects are established in other settings, operate only in the arm whose glycaemia improves, and could account for a clinically meaningful share of the gap. Weight-promoting co-therapies and differences in population composition are also plausible contributors. Diminished appetite suppression in T2D is plausible but untested. Impaired GIP signalling and intrinsic insulin-resistance-driven pharmacological resistance are the least supported explanations.

The single most important missing piece of evidence is a matched energy-balance study. It would measure energy intake, total energy expenditure and urinary energy losses in people with and without T2D receiving the same incretin agent, and formally test which components mediate the difference in weight loss.

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Declarations

Author contributions [To be completed.] **Competing interests** [To be completed.] **Funding** [To be completed.]

Methods note This is a narrative review informed by structured evidence retrieval (Supplementary Methods); it is not a systematic review and does not claim PRISMA compliance.

Supplementary information

Supplementary Methods: evidence retrieval and verification

Type: structured, targeted retrieval for a narrative review. No PRISMA screening, no duplicate independent screening, no risk-of-bias tool; not exhaustive.

Sources and tools

- PubMed/MEDLINE via PubMed MCP connector (search, metadata, PMC full text, citation matching).
- ClinicalTrials.gov via MCP connector (trial registration/eligibility; posted results not returned by the tool).
- Unpaywall API, Europe PMC, Crossref, NCBI E-utilities (enabled mid-project by network allowlisting).
- Open-access repository copies (UCL Discovery, Ulster Pure). Publisher sites (nejm.org, thelancet.com, Wiley) returned 403 to automated access.

Strategy

1. **Known-item retrieval** of landmark trials and physiology papers by citation (journal/volume/page/author), then verification of every record's abstract.
2. **Targeted searches** (examples):
    - `orforglipron[Title] AND (N Engl J Med[Journal] OR Lancet[Journal])`
    - `cagrilintide semaglutide REDEFINE`
    - `STEP-HFpEF pooled analysis semaglutide diabetes status weight`
    - `semaglutide prediabetes normoglycemia STEP trials glycemic status`
    - `tirzepatide prediabetes obesity 176 weeks diabetes prevention SURMOUNT-1`
    - `maridebart cafraglutide obesity phase 2 type 2 diabetes`
    - `(GLP-1 receptor agonist OR liraglutide OR semaglutide OR exenatide) AND energy expenditure AND (indirect calorimetry OR respiratory chamber) AND type 2 diabetes AND randomized`
    - `(GLP-1 receptor agonist ...) AND gastric emptying AND baseline gastric emptying AND type 2 diabetes AND scintigraphy`
    - `tirzepatide[Title] AND (energy intake OR appetite OR food intake) AND obesity; Martin CK[Author] AND tirzepatide`
    - Title-restricted searches for "(diabetes) AND (attenuated|less|lower) AND (weight loss)" — returned no dedicated comparative meta-analysis.
3. **Backward/forward tracking** from pivotal trials and pooled analyses (e.g., STEP-HFpEF pool, Perreault 2022).

Inclusion priorities

Matched-programme RCTs (same molecule/dose/duration/estimand) > harmonised/IPD or single-protocol cohorts > human physiology (clamps, tracers, calorimetry, ad libitum intake, fMRI) > observational cohorts > animal/cellular mechanism.

Known retrieval gaps

- Embase, Cochrane CENTRAL, conference abstracts not searched.
- Supplementary appendices of most NEJM/Lancet trials not accessible (subgroups by baseline HbA1c, background therapy, sex).
- No formal search for unpublished IPD analyses or regulatory (FDA/EMA) review documents.

Supplementary Table 1 | Structured extraction of clinical trials and cohorts

trial	agent	dose	weeks	diabetes status	n randomised	baseline bmi	baseline hba1c pct	female pct	background therapy	estimand	drug pct change	placebo pct change	etd pp	verification level	pmid
STEP 1	semaglutide s.c.	2.4 mg weekly	68	none (HbA1c<6.5% required)	1961	37.9	5.7	74.1	none	treatment policy	-14.9	-2.4	-12.4	full text (repository copy)	33567185
STEP 2	semaglutide s.c.	2.4 mg weekly (vs 1.0 mg)	68	T2D (HbA1c 7-10%)	1210	>=27 (eligibility)	7-10 (eligibility)	NR	0-3 oral agents; stratified by background therapy	treatment policy	-9.6	-3.4	-6.2	abstract	33667417
SCALE Obesity and Prediabetes	liraglutide	3.0 mg daily	56	none (61.2% prediabetes)	3731	38.3	NR	78.5	none	LOCF	-8.4 kg	-2.8 kg	-5.6 kg	abstract	26132939
SCALE Diabetes	liraglutide	3.0 mg daily (vs 1.8)	56	T2D (HbA1c 7-10%)	846	>=27	7-10	NR	0-3 OADs (metformin/TZD/SU)	NR in abstract	-6.0	-2.0	-4.0	abstract	26284720
SCALE Insulin	liraglutide	3.0 mg daily	56	T2D	396	NR	NR	NR	basal insulin + <=2 OADs; IBT	NR in abstract	-5.8	-1.5	-4.3	abstract	32139381
SURMOUNT-1	tirzepatide	5/10/15 mg weekly	72	none	2539	38.0	NR	NR	none	treatment regimen	-15.0/-19.5/-20.9	-3.1	-11.9/-16.4/-17.8 (calc)	abstract	35658024
SURMOUNT-2	tirzepatide	10/15 mg weekly	72	T2D (HbA1c 7-10%; mean 8.02; duration 8.5 y)	938	36.1	8.02	50.7	metformin 88.7%; SU 26.7%; SGLT2i 20.4%; TZD 3.5%	treatment regimen	-12.8/-14.7	-3.2	-9.6/-11.6	full text incl. supplement (repository copy)	37385275
STEP-HFpEF	semaglutide s.c.	2.4 mg weekly	52	none	529	>=30	NR	NR	HF therapy	treatment policy	-13.3	-2.6	-10.7	abstract + pooled full text	37622681
STEP-HFpEF DM	semaglutide s.c.	2.4 mg weekly	52	T2D (HbA1c<=10%)	616	>=30	NR	NR	HF therapy + glucose-lowering drugs (SU/insulin guidance)	treatment policy	-9.8	-3.4	-6.4	abstract + pooled full text	38587233
REDEFINE 1	cagrilintide+semaglutide	2.4/2.4 mg weekly	68	none	3417	NR	NR	NR	none	treatment policy	-20.4	-3.0	-17.3	abstract	40544433
REDEFINE 2	cagrilintide+semaglutide	2.4/2.4 mg weekly	68	T2D (HbA1c 7-10%)	1206	>=27	7-10	NR	NR in abstract	treatment policy	-13.7	-3.4	-10.4	abstract	40544432
ATTAIN-1	orforglipron (oral)	6/12/36 mg daily	72	none	3127	NR	NR	NR	none	treatment regimen	-7.5/-8.4/-11.2	-2.1	-5.4/-6.3/-9.1 (calc)	abstract	40960239
ATTAIN-2	orforglipron (oral)	6/12/36 mg daily	72	T2D (HbA1c 7-10; mean 8.05)	1613	35.6	8.05	46.9	NR in abstract	treatment regimen	-5.1/-7.0/-9.6	-2.5	-2.7/-4.5/-7.1	abstract	41275875
MariTide phase 2	maridebart cafraglutide	140-420 mg q4w/q8w	52	both (single protocol)	592 (465 + 127)	37.9 / 36.5	NR	63 / 42	NR	treatment policy	-12.3 to -16.2 / -8.4 to -12.3	-2.5 / -1.7	calc	abstract	40549887
retatrutide phase 2 (obesity)	retatrutide	1-12 mg weekly	48	none	338	NR	NR	48.2	none	NR	-24.2 (12 mg)	-2.1	-22.1 (calc)	abstract	37366315

trial	agent	dose	weeks	diabetes status	n randomised	baseline bmi	baseline hba1c pct	female pct	background therapy	estimand	drug pct change	placebo pct change	etd pp	verification level	pmid
retatrutide phase 2 (T2D)	retatrutide	0.5-12 mg weekly	36	T2D (HbA1c 7-10.5)	281	25-50	NR	56	diet +/- metformin	NR	-16.94 (12 mg)	-3.00	-13.9 (calc)	abstract	37385280
OASIS 1	oral semaglutide	50 mg daily	68	none	667	NR	NR	NR	none	treatment policy	-15.1	-2.4	-12.7	abstract	37385278
STEP 1/3/4 prediabetes post hoc	semaglutide s.c.	2.4 mg weekly	68	prediabetes	STEP1 subgroup 856	38.4	5.9	68.6 (sema)	lifestyle	treatment policy	-13.7 (STEP 1)	-2.4	-11.31	full text (PMC)	35724304
SURMOUNT-1 3-year prediabetes	tirzepatide	5/10/15 mg weekly	176	prediabetes	1032	NR	NR	NR	none	NR	-12.3/-18.7/-19.7	-1.3	calc	abstract	39536238
real-world cohort	semaglutide	1.7-2.4 mg weekly	26	mixed	175	41.3	NR	75.4	clinical	observed	-7.2 (T2D) vs -11.8 (no T2D) at 6 mo	none	NA	abstract	36121652
EHR cohort	tirzepatide vs semaglutide (T2D-labelled)	various	up to 52	mixed (52% T2D)	18386	39.1	NR	70.5	clinical	on-treatment	see paper (figure)	none	NA	full text (PMC)	38976257
behavioural programme	lifestyle	NA	NR	paired	24	NR	NR	50	NA	observed	-7.5 kg (T2D) vs -13.4 kg (spouses)	none	NA	abstract	3677974

Supplementary Table 2 | Mechanistic studies

source key	mechanism domain	population	model	design	measured outcome	key finding	relevance to T2D question	causal status
BLUNDELL2017	energy intake	obesity, no T2D (n=30)	human	RCT crossover 12 wk sema 1.0	ad lib intake, appetite, RMR	intake -24%; RMR unchanged	indirect (non-T2D)	causal for drug effect on intake
FRIEDRICHSEN2021	energy intake / GE	obesity, no T2D (n=72)	human	RCT 20 wk sema 2.4	ad lib intake; paracetamol GE	intake -35%; no GE delay at wk 20	indirect	causal (intake)
MARTIN2025	energy intake / reward	overweight/obesity, no T2D (n=114)	human	RCT 6 wk tirz vs lira vs pbo	ad lib intake; fMRI	-524.6 kcal; reduced food-cue activation	indirect	causal (intake)
GUTZWILLER1999	energy intake	T2D (n=12)	human	crossover IV GLP-1	ad lib intake	-27%	direct (T2D effect present)	causal
FLINT2013	energy intake	T2D (n=18)	human	crossover lira 1.8, 3 wk	ad lib intake	-18%	direct (T2D)	causal
HEISE2023	energy intake	T2D (n=117)	human	RCT 28 wk tirz/sema/pbo	appetite, ad lib intake, DXA	both reduce appetite; intake similar tirz vs sema despite more weight loss	direct (T2D)	partial
VANBLOEMENDAAL2014	CNS reward	obese T2D, obese NGT, lean (n=48)	human	crossover exenatide ± exendin 9-39, fMRI	food-cue BOLD, intake	reductions in T2D and obese; receptor-dependent	direct (both groups; magnitude not compared in abstract)	causal (receptor)
TENKULVE2015	CNS reward	T2D (n=20)	human	crossover lira vs glargine 12 wk	fMRI	effect at 10 d, not 12 wk	direct (T2D)	associational over time
FARR2016	CNS	T2D; human brain tissue	human	crossover lira; immunohistochemistry	GLP-1R localisation; fMRI	GLP-1R in hypothalamus, medulla, parietal cortex	direct (T2D)	descriptive
ROSENBAUM2022	CNS gliosis	obesity NGT/IGT/T2D (n=67)	human	cross-sectional + prospective MRI	MBH T2 relaxation	gliosis greater in IGT/T2D	direct (association)	no
SECHER2014	CNS circuits	rats/mice	animal	pharmacology, KO, electrophysiology	weight; POMC/NPY activity	ARC POMC/CART mediate lira weight loss	mechanistic only	causal in rodents
GABERY2020	CNS circuits	rodents	animal	imaging, c-Fos, transcriptomics	brain access, intake, EE	CVO access; intake down without EE fall	mechanistic only	causal in rodents
KANOSKI2011	vagal/CNS	rats	animal	antagonist, deafferentation	intake	vagal + CNS GLP-1R contribute	mechanistic only	causal in rodents
BRIERLEY2021	CNS	mice	animal	chemogenetics	intake	PPG neurons not required for GLP-1RA anorexia	mechanistic only	causal in rodents
NAUCK2011	gastric emptying	healthy (n=9)	human	IV GLP-1 8.5 h	GE	rapid tachyphylaxis	indirect	causal
UMAPATHYSIVAM2014	gastric emptying	healthy men (n=10)	human	crossover infusion regimens	scintigraphic GE	prolonged exposure attenuates slowing	indirect	causal
HALAWI2017	gastric emptying	obesity (n=40)	human	RCT lira 3.0 16 wk	GE, satiation, weight	GE delay correlates with weight loss	indirect	associational
URVA2020	gastric emptying	mice; humans with/without T2D	human+animal	phase 1, 4 wk	acetaminophen GE	tachyphylaxis; residual delay in T2D	direct (T2D)	descriptive
FRASER1990	gastric emptying	T1D (n=10)	human	clamp crossover	scintigraphic GE	hyperglycaemia slows GE	indirect (T1D)	causal (glycaemia->GE)
RAYNER2020	gastric emptying	metformin-treated T2D (n=30)	human	RCT lixisenatide 8 wk	scintigraphy, tracers	sustained slowing; effect depends on baseline GE	direct (T2D)	causal
NAUCK1993	GIP/GLP-1 islet	mild T2D vs controls	human	hyperglycaemic clamp infusions	C-peptide, glucagon	GIP effect -54%; GLP-1 preserved	direct	causal (islet)
HOJBERG2009	GIP/GLP-1 islet	poorly controlled T2D (n=8)	human	4 wk insulin near-normoglycaemia	clamp C-peptide	3-4x improved incretin response	direct	causal (glucotoxicity)
XU2007	receptor expression	rat islets	animal	Px, clamp, culture	GLP-1R/GIPR mRNA	hyperglycaemia downregulates receptors	mechanistic only	causal in rodents
BERGMANN2019	GIP appetite	overweight men, no T2D (n=17)	human	crossover infusions	ad lib intake	GIP adds nothing to GLP-1 intake effect	indirect	causal (null)
ZHANG2021	GIP CNS	mice	animal	CNS-Gipr KO	intake, weight	CNS GIPR needed for co-agonist superiority	mechanistic only	causal in rodents
ADRIAENSSENS2019	GIP CNS	mice (+ human RNAscope)	animal	chemogenetics	intake	hypothalamic Gipr cells suppress intake	mechanistic only	causal in rodents
BORNER2021	GIP emesis	mice, rats, shrews	animal	pharmacology	emesis, intake	GIPR agonism blocks GLP-1RA emesis	mechanistic only	causal in animals
KILLION2018	GIPR antagonism	mice, NHP	animal	antibody	weight	antagonism lowers weight; additive with GLP-1RA	mechanistic only	causal in animals
WILLARD2020	receptor pharmacology	cells/islets	in vitro	signalling assays	cAMP, arrestin, occupancy	tirz imbalanced (GIPR) and biased	mechanistic only	pharmacology
ASMAR2010	adipose	lean healthy (n=8)	human	AV difference + clamp	adipose flux	GIP + insulin increases TAG deposition	indirect	causal (acute)
HEISE2022	insulin sensitivity	T2D (n=117)	human	RCT clamp 28 wk	M value, disposition index	tirz improves secretion, IS, glucagon	direct (T2D)	causal (drug)
ARMSTRONG2016	insulin sensitivity	NASH	human	RCT lira 12 wk tracers/clamps	hepatic/adipose IS, DNL	improved	indirect	causal (drug)
KELLEY2000	substrate	IR humans/rodents	review	synthesis	fuel selection	metabolic inflexibility	background	n/a
HALL2015	substrate vs energy	obesity (n=19)	human	metabolic ward isocaloric	fat balance by chamber	lower insulin did not increase fat loss	indirect (principle)	causal

source key	mechanism domain	population	model	design	measured outcome	key finding	relevance to T2D question	causal status
BOGARDUS1986	glycosuria/REE	obese Pima T2D vs NGT	human	cross-sectional + 6 wk tolazamide	RMR, urine glucose, intake	RMR +5% in T2D; UGE 128→11 g/d; RMR falls	direct (T2D energetics)	causal (glycaemia->REE, UGE)
CARLSON1993	glycosuria/EE	T1D (n=6)	human	2 mo intensive insulin, constant intake, chamber	EE, glycosuria, body comp	glycosuria 428→39 mmol/d; EE -5%; +2.6 kg fat	indirect (T1D)	causal
MAKIMATTILA1999	glycosuria/REE	insulin-treated T2D (n=26)	human	RCT 12 mo insulin ± metformin	BMR, intake, glucosuria	glycaemia independent determinant of lower BMR	direct (T2D)	associational/causal
RAVE2006	renal glucose	T2D (n=22)	human	stepped clamps	UGE vs glucose	0.06→0.77 mmol/min; threshold 7.7-12.2 mM	direct (T2D)	dose-response
FERRANNINI_G2015	glycosuria energetics	T2D (n=86)	human	90 wk empagliflozin + model	weight vs glycosuric energy	206 kcal/d; intake +13% compensation	direct (T2D energetics)	model-based
POLIDORI2016	appetite feedback	T2D (n=153)	human	52 wk canagliflozin + model	calculated intake	+100 kcal/d per kg lost	direct (T2D)	model-based
LOOK2025	body composition	obesity, no T2D (n=160)	human	DXA substudy SURMOUNT-1	fat/lean	75% fat / 25% lean	indirect	descriptive
MCCRIMMON2020	body composition	T2D (n=178)	human	DXA substudy SUSTAIN 8 (no placebo)	fat/lean	fat -3.4 kg, lean -2.3 kg	direct (T2D)	descriptive
RAVUSSIN2025	EE/adaptation	obesity (no T2D) + mice	human+animal	phase 1 RCT calorimetry	EE, adaptation, RER	no adaptation change; fat oxidation up	indirect	causal (drug)
VANEYK2019	REE	T2D (n=50)	human	RCT lira 26 wk	REE, BAT	REE reduced wk 4-12	direct (T2D)	causal (drug)
HARDER2004	24h EE	obese T2D (n=33)	human	RCT lira 0.6 8 wk chamber	24h EE, DXA	no EE change	direct (T2D)	causal (null)
GOLUBIC2024	energy balance	T2D (n=28 randomised)	human	RCT cotadutide 42 d DLW + IC	EI, EE	weight loss from EI; small EE change	direct (T2D)	causal (drug)
UKPDS33	co-therapy	newly diagnosed T2D (n=3867)	human	RCT	weight	insulin +4.0 kg; SU +1.7-2.6 kg	direct (T2D)	causal (drug)
WING1987	behaviour	T2D + nondiabetic spouses (12 pairs)	human	paired behavioural programme	weight, intake	7.5 vs 13.4 kg; smaller intake reduction in T2D	direct comparison	associational

Supplementary Table 3 | Claim-to-evidence ledger

claim id	claim	source keys	source access	verification status	caveat
C01	Placebo-adjusted weight loss with sema 2.4 is -12.4 pp without diabetes vs -6.2 pp with T2D	STEP1; STEP2	abstract + STEP1 full text	verified	between-trial comparison
C02	Across 8 matched comparisons T2D/no-T2D ETD ratio median 0.62 (0.50-0.78)	Table 1 sources	derived (analysis/derived.py)	verified calculation	pairs not independent; between-trial
C03	STEP-HFpEF pooled IPD heterogeneity for weight p<0.0001, I2 95.8%; ~40% less weight loss in T2D	HFpEF_POOL	PMC full text	verified	diabetes not randomised
C04	Placebo-arm weight change not smaller in T2D trials	STEP1/2, SURMOUNT1/2, ATTAIN1/2, REDEFINE1/2, STEP-HFpEF/DM	abstracts	verified	MariTide placebo -1.7 vs -2.5 is the exception
C05	Prediabetes subgroup in STEP 1 lost 13.7% (ETD -11.31); implied normoglycaemic ~15.9%	PERREAULT2022; STEP1	PMC full text + derived	verified (derived value approximate)	weighted-mean assumption
C06	GI AEs 72.5% (REDEFINE 2) vs 79.6% (REDEFINE 1); AE discontinuations similar ATTAIN-1/2	REDEFINE1/2; ATTAIN1/2	abstracts	verified	
C07	SURMOUNT-2 efficacy-estimand ETD -10.1/-12.4	SURMOUNT2	full text	verified	
C08	Semaglutide exposure determined chiefly by body weight (T2D popPK)	CARLSSON2018; OVERGAARD2021	abstracts	verified	T2D-only data; cross-population exposure not compared
C09	Strathe model used baseline HbA1c as covariate; effect size not verified	STRATHE2023	abstract	partially verified	full text not retrievable (Wiley 403)
C10	Incretin effect 36% (T2D) vs 73% (controls)	NAUCK1986	abstract	verified	
C11	Reduced incretin effect is a consequence of the diabetic state	KNOP2007; HOJBERG2009	abstracts	verified (as authors' interpretation)	
C12	GLP-1 insulinotropic activity preserved, GIP reduced 54% in mild T2D	NAUCK1993	abstract	verified	mild T2D, HbA1c 6.3%
C13	Appetite/intake suppressed by GLP-1 agonism in T2D	GUTZWILLER1999; FLINT2013; HEISE2023; VANBLOEMENDAAL2014	abstracts	verified	
C14	No study compares appetite suppression magnitude by diabetes status under matched conditions	search (PubMed)	search negative	not found	absence-of-evidence claim; limited by search scope
C15	Tirz vs sema in T2D: intake did not differ despite greater weight loss	HEISE2023	abstract	verified	single test meal
C16	Hypothalamic gliosis greater in IGT/T2D	ROSENBAUM2022	abstract	verified	association
C17	GE slowing subject to tachyphylaxis	NAUCK2011; UMAPATHYSIVAM2014	abstracts	verified	contradicted in part by JONES2020 (exenatide QW)
C18	GIP does not augment GLP-1 intake suppression in humans	BERGMANN2019	abstract	verified	acute infusion, no T2D
C19	Attenuation ratio similar across GIPR agonist, antagonist and GLP-1-only agents	Table 1	derived	verified calculation	between-trial
C20	Isocaloric carb restriction lowers insulin but not more fat loss	HALL2015	abstract	verified	6-day ward study
C21	No human T2D study measured muscle glucose uptake, oxidation, glycogen and total EE together during incretin therapy	search (PubMed)	search negative	not found	absence-of-evidence claim
C22	Glycogen-water shifts could change weight independent of fat	general physiology	inference	unsourced hypothesis	labelled as inference in text
C23	UKPDS: insulin +4.0 kg, SU +1.7-2.6 kg	UKPDS33	abstract	verified	DOI derived from PII
C24	Lira 3.0 ETD -4.3 in insulin-treated T2D vs -4.0 without insulin	SCALE_INSULIN; SCALE_DM	abstracts	verified	IBT in SCALE Insulin
C25	UGE 0.06→0.77 mmol/min at 7.8→13.3 mM; threshold 7.7-12.2	RAVE2006	abstract	verified	
C26	Sulfonylurea: UGE 128→11 g/d; RMR fall; intake cut to hold weight	BOGARDUS1986	abstract	verified	Pima, FPG 253 mg/dL
C27	Intensive insulin: glycosuria 428→39 mmol/d; EE -118 kcal/d; +2.6 kg fat	CARLSON1993	abstract	verified	T1D
C28	Improved glycaemia lowers BMR (independent determinant)	MAKIMATTILA1999	abstract	verified	authors' conclusion
C29	SGLT2i glycosuria 206 kcal/d; observed weight 29% of predicted; intake +13%	FERRANNINI_G2015	abstract	verified	model-based
C30	100 kJ/d → ~1 kg steady state; half in ~1 y; fat 39.5 MJ/kg, lean 7.6 MJ/kg	HALL2011	PMC full text	verified	
C31	Scenario offsets ~1 to ~6.7 kg by 68-72 wk	analysis/derived.py	calculation	assumption-driven	inputs not measured in incretin trials
C32	Tirz DXA: 75% fat, 25% lean	LOOK2025	abstract	verified	no T2D
C33	Incretin agonists do not raise EE in humans	BLUNDELL2017; HARDER2004; VANEYK2019; RAVUSSIN2025; GOLUBIC2024	abstracts	verified	REE may fall
C34	T2D lost less weight than nondiabetic spouses in behavioural programme	WING1987	abstract	verified	n=12 pairs
C35	Real-world: T2D 7.2% vs 11.8% at 6 mo (sema); no-T2D lost more on both drugs (EHR)	GHUSN2022; RODRIGUEZ2024	abstract; PMC full text	verified	observational
C36	MariTide T2D cohort -8.4 to -12.3% vs obesity -12.3 to -16.2%	MARITIDE2025	abstract	verified	dose arms differ

Supplementary Note: quality control, limitations and analyses not performed

Verification performed

- **Bibliography:** all 94 cited references resolve to PubMed records with DOIs. The bibliography was generated programmatically from NCBI esummary. For UKPDS 33 the PubMed record has no DOI, so the DOI was derived from the publisher PII and is flagged as such.
- **Per-source verification:** for every source, the population, design and the specific numbers used in the text were checked against the abstract (all sources) or the full text. The log is in `references/verified_log.md`, and claims are mapped in `evidence/claim_evidence_ledger.csv`.
- **Full texts read:** STEP 1 and SURMOUNT-2 including its supplement (open-access repository copies), the STEP-HFpEF pooled analysis, Perreault 2022, Rodriguez 2024 and Hall 2011 (PMC).
- **Corrections made during audit:**
    - i. A cross-drug comparison of efficacy estimands (tirzepatide vs semaglutide) was removed.
    - ii. An overstatement that liraglutide 0.6 mg “normalised HbA1c” was corrected to “improved glycaemia”.
    - iii. An unsourced claim that agonists “reset” the appetite set point was relabelled as an untested interpretation.

Sources accessed at abstract level only (numbers limited to the abstract)

STEP 2, SURMOUNT-1, SCALE Obesity and Prediabetes, SCALE Diabetes, REDEFINE 1 and 2, ATTAIN-1 and 2, STEP-HFpEF and STEP-HFpEF DM (individually), STEP 6, OASIS 1, PIONEER PLUS, both retatrutide phase 2 trials, maridebart cafraglutide, Strathe 2023, and most physiology papers (Wing 1987, Bogardus 1986, Rave 2006, Heise 2022/2023, Ravussin 2025, Mäkimattila 1999, Carlson 1993).

Items not verified or unresolved

1. **STEP 2:** the value for the 1.0 mg arm and the baseline weight/BMI were not in the abstract and are not stated in the manuscript.
2. **Strathe 2023:** whether diabetes status or HbA1c reduced the modelled maximal weight response is not verified, because the full text was blocked by the publisher. The manuscript states this explicitly.
3. **STEP 6:** the subgroup by diabetes status is not in the abstract and was not used.
4. **Baseline characteristics (sex, age, background therapy) of several trials:** these are not reported in their abstracts, so between-programme composition differences are described only where verified (STEP 1, SURMOUNT-2, ATTAIN-2, maridebart cafraglutide, STEP-HFpEF pool).
5. **Glycogen–water contribution:** this is general-physiology inference without a cited human measurement in this context, and it is labelled as inference.
6. **Hypoglycaemia-avoidant eating** with sulfonylureas or insulin is presented as a hypothesis, without a supporting primary source.
7. **“No study has compared appetite suppression by diabetes status”** and **“no study measured muscle glucose uptake, oxidation, glycogen and total EE together”** are absence-of-evidence statements. They are limited by the scope of the search (PubMed only).

Derived values and assumptions

- **Matched-pair ratios** (`analysis/derived.py`): these are ratios of published ETDs. Pairs are not independent (two doses come from the same trials), and no meta-analytic pooling or confidence intervals were computed for the ratios.
- **STEP 1 normoglycaemic estimate (~−15.9%):** this assumes the overall mean is the subgroup-size-weighted average of the subgroup means, and it ignores differences in imputation.
- **Energy scenarios (Table 2):** glycosuria reductions of 10/25/45 g/day and REE reductions of 0/40/90 kcal/day are assumptions anchored to measured studies in other settings. The conversion uses Hall 2011 (≈23.9 kcal/day per kg at steady state; exponential approach with a 1-year half-time). The scenarios are not predictions for any trial.

Analyses not performed

- Formal meta-analysis or meta-regression of the T2D × treatment interaction.
- Risk-of-bias assessment (RoB 2) and GRADE certainty rating.
- Adjustment of matched comparisons for age, sex, BMI or background therapy (requires IPD).
- Searches of Embase, Cochrane and regulatory documents (FDA/EMA reviews may contain subgroup analyses).

Access limitations

- Publisher sites (NEJM, Lancet, Wiley, figshare) refused automated access (HTTP 403 or bot challenge) even after network allowlisting.
- Supplementary appendices for most trials were therefore unavailable.
- Supplying the PDFs of STEP 2, SURMOUNT-1, REDEFINE 1/2, ATTAIN-1/2 and the STEP-HFpEF trials would allow verification of baseline composition and subgroup analyses (for example, weight response by baseline HbA1c or sulfonylurea use). This is the most valuable next step for strengthening Section 2.5.

Data and code availability

All derived values (matched-pair ratios, the implied normoglycaemic estimate and energy-balance scenarios) are generated by `analysis/derived.py`; figures by `figures/make_figures.py`; numbering and bibliography by `analysis/build_manuscript.py`. Bibliographic metadata were retrieved from PubMed (NCBI E-utilities).